



PINKBALLOON 2026.1

PSUM

LUI SA RIBEIRO

PROBLEMA MOTIVADOR

Static Range Sum Queries - CSES

Static Range Sum Queries

TASK | SUBMIT | RESULTS | ANALYSIS | STATISTICS | TESTS | QUEUE

Time limit: 1.00 s Memory limit: 512 MB

Given an array of n integers, your task is to process q queries of the form: what is the sum of values in range $[a, b]$?

Input

The first input line has two integers n and q : the number of values and queries.

The second line has n integers $x_1, x_2, \dots, x_n$: the array values.

Finally, there are q lines describing the queries. Each line has two integers a and b : what is the sum of values in range $[a, b]$?

Output

Print the result of each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Range Queries

Static Range Sum Queries	
Static Range Minimum Queries	
Dynamic Range Sum Queries	
Dynamic Range Minimum Queries	
Range Xor Queries	
Range Update Queries	
Forest Queries	
Hotel Queries	
...	

Your submissions

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Output

Print the result of each query.

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq x_i \leq 10^9$
- $1 \leq a \leq b \leq n$

Example

Input:

```
8 4
3 2 4 5 1 1 5 3
2 4
5 6
1 8
3 3
```

Output:

```
11
2
24
4
```

Your submission

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**Como saber a soma dos números de
um subarray ?**

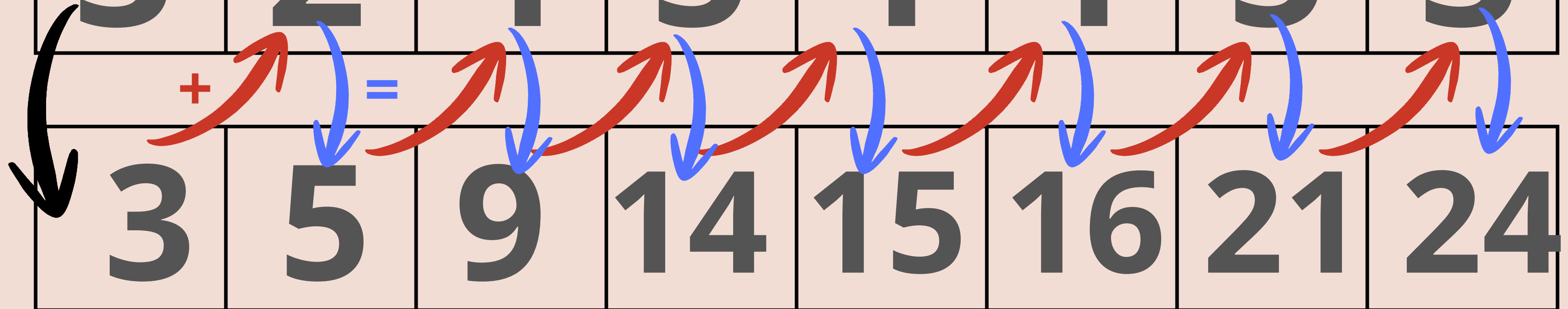
PROBLEMA MOTIVADOR

Static Range Sum Queries - CSES

- Armazenar a soma de prefixos no vetor Psum
- Última posição do vetor psum armazena a soma de todos os N valores armazenados no vetor;
- Soma de um intervalo $[l, r]$: $\text{psum}[r] - \text{psum}[l-1]$

vetor v

3	2	4	5	1	1	5	3
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vetor psum

$$\text{psum}[i] = \text{psum}[i-1] + v[i]$$

ARRAY DE DIFERENÇAS

- Representar mudanças entre elementos consecutivos
- Em vez de armazenar os valores, armazena as diferenças

**Obrigada
pela
atenção.**

